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Hyperbolic smoothing and penalty techniques applied to molecular structure determination

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ABSTRACT

This work considers the problem of estimating the relative positions of all atoms of a protein, given a subset of all the pair-wise distances between the atoms. This problem is NP-hard, and the usual formulations are nonsmoothed and nonconvex, having a high number of local minima. Our contribution is an efficient method that combines the hyperbolic smoothing and the penalty techniques that are useful in obtaining differentiability and reducing the number of local minima.

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1. Introduction

The problem of determining molecular structures has attracted great interest due to its application in relevant areas such as medicine, pharmacy, biology, design of materials, and chemistry [25]. Until 1984, the X-ray crystallography technique used to be the ultimate tool for obtaining information about protein structure at atomic resolution. The introduction of nuclear magnetic resonance (NMR) as a technique for protein structure determination made it possible to obtain structures with high precision in an environment (solution) that is much closer to the natural surroundings of a living organism than the crystals used in crystallography [25].

NMR experiments yield valuable information: a network of bounds for distances involving pairs of atoms spatially close, at distances less than 5–6 Å (1 Å $= 10^{-10}$ m). In the *molecular distance geometry problem* (MDGP), the challenge is to obtain a valid three-dimensional configuration for the atoms, i.e., a configuration that is compatible with the network of bounds obtained in the NMR experiment. The MDGP can be precisely defined as

$$\begin{aligned} &\text{determine } x_1, x_2, \ldots, x_m \\ &\text{subject to } l_{ij} \le \|x_i - x_j\| \le u_{ij}, \quad \forall (i,j) \in K, \\ &\qquad\qquad x_i \in \mathbb{R}^3, \ i = 1, \ldots, m, \end{aligned}$$

where $K \subset \{1, \ldots, m\} \times \{1, \ldots, m\}$ is a set that identifies the available lower (l_{ij}) and upper (u_{ij}) bounds for the pair-wise Euclidean distances, and m is the number of atoms in the protein.

The MDGP can be reformulated as a mathematical programming problem.

Definition 1 (*MDGP*).

$$\begin{aligned} (P) \ \min f(x) = &\sum_{(i,j)\in K} \max\{l_{ij} - \|x_i - x_j\|, 0\} \\ &+ \max\{\|x_i - x_j\| - u_{ij}, 0\} \\ &\text{where } x = (x_1, \ldots, x_m) \in \mathbb{R}^{3m}, \\ &x_k \in \mathbb{R}^3, \ \forall k \in \{1, \ldots, m\}. \end{aligned} \tag{1}$$

It is easy to see that $f(x_1, \ldots, x_m) = 0$ if and only if all the restrictions $l_{ij} \le \|x_i - x_j\| \le u_{ij}$ are satisfied.

The contribution of this work is a new algorithm for the MDGP. Our proposal combines the hyperbolic smoothing and penalty techniques [33]. This approach allows the application of classical optimization methods by introducing differentiability into formulation (1) and, more importantly, it generates trajectories that have a great chance to converge to the global minimum of (P).

In Section 2, we point out the complexity of the MDGP and the different approaches found in the literature. In Section 3, we propose a new algorithm to solve the MDGP by using hyperbolic functions. Section 4 presents computational experiments carried out with instances generated from real proteins. Finally, in Section 5, we summarize the contributions of this work.

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2. The MDGP

The MDPG can be posed as a nonlinear global optimization problem, such as (P), for instance. Unfortunately, the problem of finding the ideal formulation for the MDGP remains unsolved [7,22,24]. There are no results of convexity. Besides, in general, the formulations using penalty functions are not differentiable and have many local minima, which precludes the direct application of classical and more robust optimization methods. The nondifferentiability arises from the inclusion of Euclidean norm in the definition of the constraints and also from the use of the $\max\{\cdot, 0\}$ as the penalty function. The large number of local minima is connected to the strong combinatorial appeal of the MDGP, which gets worse with the increasing number of atoms considered [8,15,18].

Particular cases of the MDGP are solved in a relatively easy way. For instance, when we know all distances $d_{ij} = \|x_i - x_j\|$, i.e., $d_{ij} = l_{ij} = u_{ij}$ and $K = \{1, 2, \ldots, m\}^2$, a solution can be obtained by factoring the matrix $D = [d_{ij}]$. In fact, a very simple algorithm presented by Dong and Wu generates a solution to the MDGP in linear time, in the case where all distances are known [8].

However, in practice, the NMR experiments just provide the subset of the distances between atoms that are spatially close [25], and the data accuracy is limited. Thus, in a real scenario, the set K is sparse, and $l_{ij} < u_{ij}$. This form of the MDGP seems to be easier to solve, since the constraints are less restrictive. However, in practice, the lower and upper limits are close, and thus the problem is still difficult to solve. In [21], Moré and Wu showed that if the upper and lower bounds are close, then the MDGP with relaxed distances belongs to the NP-hard class.

Different approaches to the MDGP have been explored: smoothing techniques [19,22], semidefinite programming [5], differences of convex functions [3], geometric inequalities [7], graph rigidity [15], geometric build up algorithms [6,31,32], alternating projections [13], multi-scaling [16,27], stochastic perturbation [37] and branch-and-prune [18] (for a review of the MDGP methods, see [17]).

3. The hyperbolic functions

The most common penalty function used in the MDGP is the function

$$p_\lambda(y) = \lambda \max\{y, 0\},$$

where λ is a parameter that indicates the penalty intensity. We alternatively propose the use of the penalty function

$$\phi_{\lambda,\tau}(y) = \lambda y + \sqrt{\lambda^2 y^2 + \tau^2},$$

where $\lambda > 0$, $\tau > 0$ and $y \in \mathbb{R}$.

Note that $\lim_{\tau\to 0} \phi_{1/2,\tau}(y) = \max\{y, 0\}$. In other words, if $\lambda = 1/2$ and τ is small, the function $\phi_{\lambda,\tau}$ is a good approximation to $\max\{\cdot, 0\}$. In Fig. 1, we see that the parameter τ controls the smoothness degree and λ controls the intensity (weight) of the penalty function $\phi_{\lambda,\tau}$.

Finally, in order to obtain the differentiability in problem (1), we propose to replace the function $\|\cdot\|$ by the hyperbolic smoothing function

$$\theta_\tau(x_i) = \sqrt{\tau^2 + \sum_{k=1}^{3} x_{ik}^2},$$

where $x_i = (x_{i1}, x_{i2}, x_{i3}) \in \mathbb{R}^3$, $\tau > 0$. It needs to be pointed out that the hyperbolic smoothing θ_τ of the Euclidean norm is a natural approach that has been frequently used not only in location theory [11,12,30] but also in other fields such as geomechanics [1,36], PDE theory [9,29], and packing problems [35].

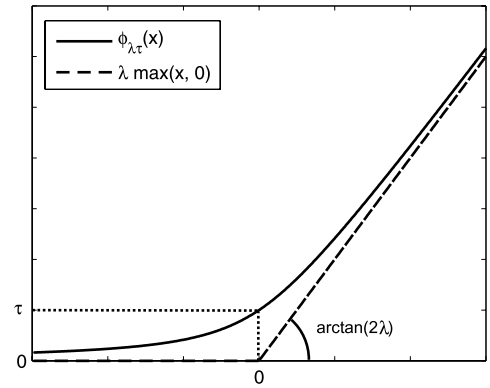

Fig. 1. In the graph of the penalty function $\phi_{\lambda,\tau}$, the parameter λ controls the intensity (slope) of penalty, and τ controls the smoothness.

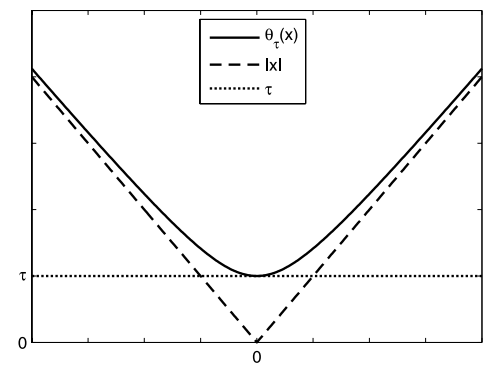

Fig. 2. The graph of the hyperbolic smoothing $\theta_\tau(x)$ is an equilateral hyperbola.

In Fig. 2, we can see clearly that the maximum distance between the original function and the smooth function θ_τ is equal to the parameter τ. The graphs of θ_τ and $\phi_{\lambda,\tau}$ are equilateral hyperbolas, which motivates the inclusion of the hyperbolic term used in the nomenclature of these functions.

Inserting the functions $\theta_\tau^{ij}(x) = \theta_\tau(x_i - x_j)$ and $\phi_{\lambda,\tau}$ in problem (1), we obtain the smoothed problem:

$$(P_{\lambda,\tau})\ \min f_{\lambda,\tau}(x) = \sum_{(i,j)\in K} \phi_{\lambda,\tau}(l_{ij} - \theta_\tau^{ij}(x)) + \phi_{\lambda,\tau}(\theta_\tau^{ij}(x) - u_{ij}),$$
$$\text{where } K \subset \{1, \ldots, m\}^2, x = (x_1, \ldots, x_m), x_k \in \mathbb{R}^3,$$
$$\forall k = 1, \ldots, m. \tag{2}$$

Problem $(P_{\lambda,\tau})$ is infinitely differentiable with respect to x and therefore allows the application of classical optimization methods. However, it should be noted that problem $(P_{\lambda,\tau})$ is not exactly equal to problem (1). Our approach to solving (P) is similar to that of the homotopy continuation methods, where the idea is to start by solving the easiest problem $(P_{\lambda,\tau})$, and then the most difficult one (P) [2,28]. We propose to solve an infinite sequence of smooth problems (P_{λ,τ_k}) that are parameterized by a descending sequence of parameters τ_k, $k = 1, 2, \ldots$, tending to zero, i.e., $\tau^{k+1} < \tau_k$ with $\lim_{k\to+\infty} \tau_k = 0$. Through this procedure, the sequence of smooth problems (P_{λ,τ_k}) gradually converges to the original problem.

Note that in the definition of the sequence (P_{λ,τ_k}), we use the same parameter λ in all terms of the objective function. This simply reflects our choice about not privileging any particular constraint. However, in an application with data from different sources, we may wish to influence the order in which the constraints are satisfied, and this could be done by setting different values for the parameter λ in each constraint.

3.1. Convex properties

Note that problem (2) is nonconvex, and the classical methods might stagnate at local optima. We claim that the use of hyperbolic smoothing and penalty functions is not justified solely by the introduction of differentiability. The great advantage in using these functions is related to the convexity, as originally pointed out by Xavier in the context of the MDGP in [34] and explored by Macambira in [23]. The following results prove that problem $(P_{\lambda,\tau})$ can be made convex by taking a smoothing parameter value τ sufficiently large.

Proposition 1. *The function $\theta_\tau^{ij} : \mathbb{R}^{3m} \to \mathbb{R}$, given by $\theta_\tau^{ij}(x) = \theta_\tau(x_i - x_j)$, is convex for all $\tau > 0$.*

Proof. It can be checked directly that the Hessian matrix of θ_τ^{ij} has the following structure:

$$\nabla^2\theta_\tau^{ij}(x) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & H_\tau^{ij}(x) & 0 & -H_\tau^{ij}(x) & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -H_\tau^{ij}(x) & 0 & H_\tau^{ij}(x) & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{3m\times 3m},$$

where

$$H_\tau^{ij}(x) = \frac{I_{3\times 3}}{\theta_\tau^{ij}(x)} - \frac{(x_i - x_j)(x_i - x_j)^T}{(\theta_\tau^{ij}(x))^3},$$

$I_{3\times 3}$ is the identity matrix, and the zeros in the matrix $\nabla^2\theta_\tau^{ij}(x)$ are in fact zero block matrices.

For any $z \neq 0$ in $\mathbb{R}^3$, it follows that

$$z^T H_\tau^{ij}(x) z \geq \frac{\|z\|^2}{\theta_\tau^{ij}(x)} - \frac{\|z\|^2\|x_i - x_j\|^2}{(\theta_\tau^{ij}(x))^3} > 0.$$

Thus, we have $H_\tau^{ij}(x) \succeq 0$.

Finally, for any $v = (v_1, v_2, \ldots, v_m) \neq 0$ in $\mathbb{R}^{3m}$,

$$v^T\nabla^2\theta_\tau^{ij}(x)v = (v_i - v_j)^T H_\tau^{ij}(x)(v_i - v_j) \geq 0.$$

Therefore, $\nabla^2\theta_\tau^{ij}(x) \succeq 0$ and, consequently, θ_τ^{ij} is a convex function. □

Corollary 1. *The objective function $f_{\lambda,\tau}$ given in (2) is convex for all $\tau > \max\{u_{ij} : (i,j) \in K\}$.*

Proof. By definition, each term of the objective function of problem $(P_{\lambda,\tau})$ is given by

$$\begin{aligned} f_{\lambda,\tau}^{ij}(x) &= \phi_{\lambda,\tau}(l_{ij} - \theta_\tau^{ij}(x)) + \phi_{\lambda,\tau}(\theta_\tau^{ij}(x) - u_{ij}) \\ &= (l_{ij} - u_{ij}) + \sqrt{\lambda^2(\theta_\tau^{ij}(x) - l_{ij})^2 + \tau^2} \\ &\quad + \sqrt{\lambda^2(\theta_\tau^{ij}(x) - u_{ij})^2 + \tau^2}. \end{aligned}$$

Note that we can rewrite $f_{\lambda,\tau}^{ij}(x)$ as

$$f_{\lambda,\tau}^{ij}(x) = (l_{ij} - u_{ij}) + h_{\lambda,\tau}^{ij}(\theta_\tau^{ij}(x), l_{ij}) + h_{\lambda,\tau}^{ij}(\theta_\tau^{ij}(x), u_{ij}),$$

where $h_{\lambda,\tau}^{ij}(x,\alpha) = \sqrt{\lambda^2(\theta_\tau^{ij}(x) - \alpha)^2 + \tau^2}$.

First, we shall prove that the function $h_{\lambda,\tau}^{ij}(x,\alpha)$ is convex if $\alpha \leq \tau$. In fact, if $\alpha \leq \tau$, we have

$$\begin{aligned} \nabla_x^2 h_{\lambda,\tau}^{ij}(x,\alpha) &= \frac{\lambda^2\tau^2\nabla\theta_\tau^{ij}(x)\nabla^t\theta_\tau^{ij}(x)}{h_{\lambda,\tau}^3(x,\alpha)} + \frac{(\theta_\tau^{ij}(x) - \alpha)\nabla^2\theta_\tau^{ij}(x)}{h_{\lambda,\tau}(x,\alpha)} \\ &\succeq 0, \end{aligned}$$

because $\alpha \leq \tau \leq \theta_\tau^{ij}(x), \forall x \in \mathbb{R}^{3m}$, and, by Proposition 1, $\nabla^2\theta_\tau^{ij}(x) \succeq 0$.

Finally, by hypothesis, $\tau > \max\{u_{ij} : (i,j) \in K\} \geq \max\{l_{ij} : (i,j) \in K\}$; thus each term of the objective function of problem $(P_{\lambda,\tau})$ is a convex function, which concludes the proof. □

In practice, the convexification of $(P_{\lambda,\tau})$ obtained by choosing high values for the parameter τ should be avoided, because the objective function $f_{\lambda,\tau}$ tends to the constant $2\tau\sqrt{\lambda^2 + 1}$ as τ grows. Thus, the parameter τ should express a harmonious compromise between problem simplification and solution consistency. That is, we should make the terms $f_{\lambda,\tau}^{ij}$ convex by choosing an appropriate value for the parameter τ, but avoiding excessively high values.

The properties of the hyperbolic smoothing and penalty techniques are used in Algorithm 1 for solving problem (P), Definition 1. Following the notation proposed in [22], we denote by **locmin** $(f, x, \mathcal{M})$ the minimizer of the function f generated by a local minimization algorithm $\mathcal{M}$, taking x as the starting point.

Despite the theory presented, it needs to be pointed out that the routine *sph* is heuristic. This means that there is no guarantee that a global minimum will be reached for all possible starting points. Part of this uncertainty is strongly related to the lack of a precise way of setting the smoothing parameter τ and also to the application of the local minimization procedures at each step.

Algorithm 1 Routine *sph*

```
sph(x, λ, τ, ρ) : ρ ∈ (0, 1);
  while(f_{λ,τ}(x) > ε_f) do
    x = locmin(f_{λ,τ}, x, M);
    τ = ρτ;
  end do
return (x, f_{λ,τ}(x));
```

4. Computational experiments

Our procedure *sph* was implemented in *C*, except for local minimization routine *va*35, encoded in FORTRAN, and available from the Harwell Subroutine Library. The routine *va*35 implements the method BFGS with limited memory [20] (For additional information on this routine, see http://www.hsl.rl.ac.uk/catalog.html). We used a computer with an Intel Core 2 Duo CPU T5670 1.80 GHz, 3.0 GB of RAM, and Linux OS-64 bit.

The instances considered were derived from the three-dimensional structure of the fragments made up of the first 100 and 200 atoms of the chain *A* of the protein 1 GPV [14,26]. For each fragment, we generated a set of constraints considering only atoms in the same residue or the neighboring residues. Formally,

$$K = \{(i,j) : x_i \in R(k), x_j \in R(k) \cup R(k+1)\},$$

where $R(k)$ represents the k-th residue.

As Moré and Wu in [22], we consider as a solution any set of coordinates $x \in \mathbb{R}^{3m}$ such that

$$(1 - \tau_d)l_{ij} \leq \|x_i - x_j\| \leq (1 + \tau_d)u_{ij}, \quad (i,j) \in K, \tag{3}$$

where $\tau_d = 10^{-2}$. This tolerance reflects the precision for the bond lengths [10,22].

The bounds l_{ij} and u_{ij} were given by the equations

$$l_{ij} = (1 - \varepsilon_b)\|\hat{x}_i - \hat{x}_j\|, \qquad u_{ij} = (1 + \varepsilon_b)\|\hat{x}_i - \hat{x}_j\|,$$

where $\hat{x} = (\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_m)$ represents the known structure of protein *1GP* available in the PDB (Protein Data Bank [4]). We generated four instances for each fragment by taking ϵ_b equal to 0.04, 0.08, 0.12, and 0.16.

The parameters of the routine *sph* were set to $\lambda = 0.5$ and $\rho = 0.99$, and the initial value of the smoothing parameter τ was defined at run time as the third quartile of the set $\{(l_{ij} + u_{ij})/2 :$

Table 1
The number of solutions (*ns*) obtained by the routines *va*35, *dgsol* and *sph* and the average CPU time (*cput*) in seconds.

ε	*va*35		*dgsol*		*sph*	
	ns	cput	ns	cput	ns	cput
100 atoms						
0.04	0	0.26	73	7.33	100	1.58
0.08	0	0.26	72	8.13	100	1.46
0.12	3	0.26	100	8.17	100	1.40
0.16	27	0.23	100	9.28	100	1.32
200 atoms						
0.04	0	0.54	44	29.25	83	8.48
0.08	0	0.54	36	30.36	88	7.06
0.12	0	0.54	100	32.57	100	6.50
0.16	7	0.51	100	33.03	100	6.01

$(i, j) \in K\}$. With this choice of the value of the parameter τ, by Corollary 1, about 75% of the terms of the objective function of the initial problem $(P_{\lambda,\tau})$ are convex. The stopped parameter λ_f of *sph*, Algorithm 1, were set to $\lambda_f = \lambda_d = 10^{-2}$.

The starting points of our experiments were generated by the procedure *struct* proposed by Moré and Wu in [22], which is described in Algorithm 2 (the source code of the routine *struct* is available at http://www.mcs.anl.gov/~more/dgsol).

Algorithm 2 Routine *struct*

```
struct(K) : K ⊂ {1, 2, ..., m}², x ∈ ℝ^{3m};
  set x = (x_1, ..., x_m) = 0;
  L = {1, 2, ..., m};
  while(L is not empty) do
    choose i ∈ L;
    set M_i = {j : (i, j) ∈ K, j ∈ L};
    for each j ∈ M_i
      generate y ∈ ℝ³ such that ‖x_i − y‖ = δ_ij;
      set x_j = y;
    end for
    remove i from L;
  end do
return x = (x_1, ..., x_m);
```

This routine generates starting points $x = (x_1, \ldots, x_m)$ that satisfy at least $m - 1$ constraints $\|x_i - x_j\| = \delta_{ij}$, where m is the number of atoms and $\delta_{ij} = (l_{ij} + u_{ij})/2$.

In the first experiment, we compared the performance of the routine *sph* with *va*35 and with the routine *dgsol* proposed by Moré and Wu in [22]. We decided to compare our approach with that of the *dgsol*, whose importance is justified in the MDGP literature, because it also uses a smoothing technique and the instances used to test the algorithm are clearly explained in [22]. Besides that, its code is available on the Internet (http://www.mcs.anl.gov/~more/dgsol). Therefore, a fair comparison between the methods could be made.

In each test, we generated 100 random points using the routine *struct*. The three routines *sph*, *va*35, and *dgsol* were initialized with these 100 random starting points. The results appear in Table 1, where *ns* and *cput* represent the number of solutions and the average CPU time computed in seconds, respectively.

Table 1 shows that the routine *sph* found more global minima than the routines *dgsol* and *va*35. The routine *sph* also demanded lower computational effort (CPU time) than the one required by the routine *dgsol*, where multidimensional integrals need to be evaluated in the calculation of the objective function and its derivatives (see [22] for more details).

In the second experiment, we verified computationally the relation between the number of local minimizers of the smoothed function $f_{\lambda,\tau}$ and the value of the parameter τ. Following the strategy adopted by Moré and Wu, we estimated the number of local minimizers of $f_{\lambda,\tau}$ for each τ by taking 100 random starting points and by minimizing $f_{\lambda,\tau}$ using the routine *va*35.

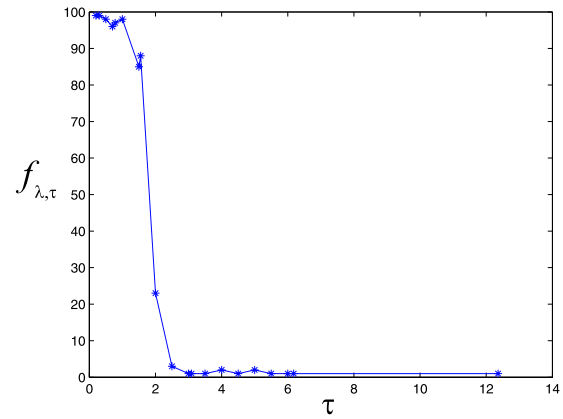

Fig. 3. The number of local minimizers of $f_{\lambda,\tau}$ as a function of τ for $\varepsilon = 0.04$ and 100 different starting points.

The number of distinct local minimizers found by *va*35 is plotted in Fig. 3. For these results, minimizers u and v of $f_{\lambda,\tau}$ are declared to be the same if

$$|f_{\lambda,\tau}(u) - f_{\lambda,\tau}(v)| \le \tau_r \max\{|f_{\lambda,\tau}(u)|, |f_{\lambda,\tau}(v)|\},$$

where $\tau_r = 10^{-6}$, or if

$$\max\{f_{\lambda,\tau}(u), f_{\lambda,\tau}(v)\} \le \tau_a, \quad \text{where } \tau_a = 10^{-2}.$$

The results in Fig. 3 show that the number of local minimizers of $f_{\lambda,\tau}$ decreases as τ increases, in accordance with the theory presented.

5. Conclusion and proposals for future work

The problem of determining molecular structures of proteins has attracted great interest due to its application in relevant areas such as medicine, pharmacy, biology, design of materials, and chemistry. The alternatives that have been used in this endeavor are experiments involving nuclear magnetic resonance (NMR), where the distances between some of the atoms of the protein can be estimated. In the molecular distance geometry problem (MDGP), the goal is to determine the three-dimensional conformation of a protein, based on estimates of the distances provided by NMR experiments.

We have presented an algorithm for the MDGP in its most realistic version, i.e., with a sparse set of inaccurate distances. Our proposal, the algorithm *sph*, starts with a formulation of least squares and tries to convexify the associated objective function through hyperbolic smoothing and penalty functions. Our approach was based on the proved property which states that appropriate values for the parameters of our smoothing function generate a convex problem.

We performed computational experiments with real data obtained from the PDB. The results indicate that the algorithm *sph* is more efficient than the multistart approach and the Gaussian smoothing technique proposed in the literature.

As a proposal for future research, we are interested in improving the MDGP formulation. In this work, we consider only distance constraints. However, by inserting restrictions arising from theoretical models of protein conformation, we can, in theory, get more significant results. Another point that deserves attention is the use of more realistic instances. For that purpose, we intend to establish partnerships with laboratories that produce real data from NMR spectroscopy.

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